Interactive Learning tool for Dyslexia Kids

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Dyslexia is a prevalent neurodevelopmental learning disability affecting 5–10% of the school-age population worldwide, characterized by significant difficulties in phonological processing, reading, spelling and word recognition. In Sri Lanka, dyslexic children face compounded challenges due to a bilingual educational environment where both Sinhala and English are used as mediums of instruction. Sri Lanka currently lacks any digitally delivered, culturally localized tool that addresses phonological awareness and spelling skills for dyslexic children in both Sinhala and English. This research identifies a critical gap: the absence of an audio-first, bilingual phonological awareness and spelling intervention tool that accommodates Sinhala-English phonological differences and supports children with severe reading difficulties. The proposed Phonological Awareness & Spelling Module addresses this gap through four core components: (1) a drag-and-drop sound block interface for letter-tile and word construction, (2) "change one sound" puzzles with integrated audio feedback, (3) spelling construction activities with minimal text dependency, (4) an error pattern detection engine for consonant reversal confusion (b/d, p/q). The module's audio-heavy, text-light design paradigm ensures children with severe reading difficulties can engage with phonological practice without first needing to read instructions. Core technologies include React Native for cross-platform mobile delivery, TensorFlow Lite for on-device consonant-confusion pattern detection, Web Speech API for audio feedback, and a custom Sinhala-English phoneme mapping engine. Evaluation will use pre/post assessments on phonological awareness sub-skills and real-time error-pattern analytics. The expected outcomes include measurable improvements in phoneme segmentation, blending, and spelling accuracy among dyslexic children aged 6–12 in Sri Lanka. The module contributes to inclusive education (SDG 4) by providing culturally and linguistically contextualized dyslexia support.

Keywords: Dyslexia, Phonological Awareness, Sinhala-English Bilingual, Spelling Intervention, Audio-First Design, Consonant Reversal Detection, Mobile Learning

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I am sincerely thankful to the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic environment, research resources and institutional support that have made this project possible.

I extend my appreciation to external supervisor Dr. Kamalani Wanigasinghe who generously shared her professional expertise regarding phonological intervention practices in Sri Lankan primary schools. Her insights have directly shaped the Phonological Awareness and Spelling Module.

Finally, this work is dedicated to every dyslexic child in Sri Lanka who has experienced the frustration of struggling with sounds and spelling in a system not yet fully equipped to support them. It is my hope that this module will meaningfully reduce those barriers.

TABLE OF CONTENT

Contents

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT 5
LIST OF TABLES 7
LIST OF ABBREVIATIONS 8
1 INTRODUCTION 9
1.1 Background and Context 9
1.2 Literature Review 10
1.3 Research Gap 12
2 OBJECTIVE 14
2.1 Main Objective 14
2.2 Specific Objectives 14
3 METHODOLOGY 15
3.1 Research Approach 15
3.2 Technical Methods and Algorithms 16
3.3 Tools and Platforms 16
3.4 Data Collection and Ethical Considerations 17
3.5 Validation Strategy and Performance Metrics 17
4 HIGH-LEVEL SYSTEM ARCHITECTURE 18
5 USER REQUIREMENTS 20
5.1 Stakeholder Identification 20
5.2 Functional Requirements 20
5.3 Non-Functional Requirements 21
6 COMMERCIALIZATION PLAN 22
6.1 Target Market and Customer Persona 22
6.2 Value Proposition 22
6.3 Revenue Model 22
6.4 Cost Estimation 22
6.5 Pricing Strategy 22
6.6 Competitive Advantage 22

6.7 Intellectual Property Considerations 23
7 BUDGET AND JUSTIFICATION 24
8 WORK BREAKDOWN STRUCTURE (WBS) 25
9 GANTT CHART 26
10 Reference 27
References 27
11 APPENDICES 29
Appendix A: Sample Questions 29
Appendix B: Preliminary UI Mockup Description 29
Appendix C: Risk Analysis 29
Appendix D: Ethical Clearance 30

LIST OF TABLES

Table 1: Comparison of Existing Dyslexia Support Tools

Table 2: Functional Requirements

Table 3: Non-Functional Requirements

Table 4: Budget Breakdown

Table 5: Work Breakdown Structure

LIST OF ABBREVIATIONS

IT	Information Technology
AI	Artificial Intelligence
SLIIT	Sri Lanka Institute of Information Technology
ML	Machine Learning
TAF	Topic Assessment Form
SDG	Sustainable Development Goal
NLP	Natural Language Processing
TTS	Text-to-Speech
UI	User Interface
UX	User Experience

1 INTRODUCTION

1.1 Background and Context

Dyslexia is one of the most common learning difficulties in the world. It makes reading, spelling and understanding the sounds in words very hard, even for children who are clever and do well in other areas. Studies show that around 10% of children worldwide are affected by dyslexia [1] [2]. In Sri Lanka, this is a real problem too, especially for children in Grades 1 to 3 who are just starting to learn how to read and write. Children with dyslexia often struggle because of phonological awareness. This is the ability to hear and play with the sounds inside words. Research has shown clearly that phonological awareness is one of the biggest factors in whether a child will struggle with reading [3]. The problem is even harder for children in Sri Lanka because they grow up speaking and hearing both Sinhala and English languages. Many tools that exist today to help children with dyslexia were made only for English-speaking children in Western countries. These tools use a lot of written text in their instructions, which is a big barrier for a child who already finds reading difficult. There are no known tools that connect the sounds of Sinhala and English to help bilingual children learn more easily.

If dyslexia is not identified and helped early, children can fall behind in school and lose confidence in themselves. Early intervention programs have been shown to greatly reduce the long-term effects of dyslexia [4]. This makes it very important to build support tools that are fun, easy to use, and work well for young Sri Lankan children. This module is part of a larger AI-based dyslexia support system being developed for Grade 1 to 3 children in Sri Lanka. The Phonological Awareness and Spelling Module specifically focuses on helping children practice the sounds in words and build spelling skills using audio-heavy, game-like activities that do not depend on reading instructions. This work is aligned with United Nations Sustainable Development Goal 4 (Quality Education), which aims to make sure that all children get a fair chance at a good education. It is also aligned with the SLIIT research cluster on AI and intelligent systems, as this module uses machine learning to detect and respond to a child's specific error patterns.

1.2 Literature Review

A number of studies have looked at how technology and machine learning can help children with dyslexia. This review covers the key works that are most relevant to this module.

Ghadekhar, Kumar, Joshi, Mishra, and Vishwakarma (2021) looked at using machine learning to predict dyslexia in children based on their phonological skills, vocabulary, memory, speed, and visual discrimination. They tested several algorithms and found that a Support Vector Machine (SVM) using Grid Search gave the best accuracy of 97.42% [1]. This work is directly useful because it shows that quiz-based and skill-based data can be used to predict and track dyslexia with high accuracy. However, the study was done in India and does not account for bilingual children who use both Sinhala and English.

Biize, Qbadou, and Mansouri (2025) studied how phonological awareness affects vocabulary learning in dyslexic children in Morocco, a country that also has children growing up with more than one language. They found that dyslexic children scored much lower on phonological awareness tests compared to children without dyslexia. They also suggested that an artificial neural network (ANN) trained on local data could help create better diagnostic tools for children in multilingual settings [2]. This study is important because it shows that tools made for one language context may not work well in another, which is the same challenge faced in Sri Lanka.

Phatangare, Jain, Chakrabarty, Chougule, and Bisen (2025) built an AI-powered system called LexiEase AI that uses models like DistilBERT and DeepSpeech to test children for dyslexia. Their system includes a phonological awareness test, a memory test, and a reading fluency test. Based on the scores, children are given a personalised learning path with reading passages, comprehension activities, memory games, and phonological improvement games [3]. The system achieved good accuracy across all three types of tests. This is a strong example of a full-featured dyslexia support system, and it shows how different types of tasks can be combined to help children. However, the system is built only for English and does not address the needs of bilingual children.

Facoetti, Franceschini, Gaggi, Galiazzo, Gori, Palazzi, and Ruffino (2014) took a different approach and built serious games for preschool children to identify early signs of dyslexia. Their games trained visual attention and sound identification skills using a multiplatform approach so the games could be used on tablets and web browsers. The games were designed to be fun and engaging so that children would want to keep playing [4]. This is helpful because it shows that game-based activities can be effective for very young children and that multiplatform support is important for reaching children in different settings.

Jency K, Lavaniya T, and Swaminathan K (2025) developed a predictive model for dyslexia that used a two-stage quiz system supervised by parents. In the first stage, children answered general questions about their language and reading habits. Those who seemed at risk were then

given a second, more detailed test. The system used Random Forest and SVM classifiers and found that both worked well, with SVM giving the highest accuracy. The study also highlighted the importance of including parental input in the assessment process [5]. This work is useful because it shows a step-by-step approach to identifying dyslexia that is practical for families.

Thelijjagoda, Chandrasiri, Hewathudalla, Ranasinghe, and Wickramanayake (2019) built a mobile application called 'The Hope' to help children with dyslexia improve their reading, writing, and speaking skills. The app used voice recognition, image processing, and gesture detection to interact with children. It was tested with children at Lady Ridgway Hospital in Sri Lanka, which is one of the very few dyslexia-related projects built specifically for Sri Lankan children [6]. This work is important because it shows that such a tool can be built and tested locally. However, it does not focus on phonological awareness training or bilingual sound mapping.

Across all these studies, a few clear gaps can be seen. No tool has been built to connect the sounds of Sinhala and English for bilingual children. Most tools rely on written text for instructions, which is a problem for children who already struggle to read. Very few systems can track specific letter confusion errors like mixing up b and d, or p and q, and give targeted help based on those errors.

Author(s) & Year	Focus	Key Finding	Gap / Limitation
Ghadekhar et al. (2021)	ML prediction of dyslexia using phonological skills, vocabulary, memory, speed, visual discrimination	97.42% accuracy with SVM — quiz-based data can reliably predict dyslexia	Does not account for bilingual children (e.g., Sinhala/English)
Biize, Qbadou & Mansouri (2025)	Phonological awareness and vocabulary learning in dyslexic children	Dyslexic children scored significantly lower; tools must be localized for multilingual settings	No tool built yet; highlights challenge of language-specific tool design
Phatangare et al. (2025)	AI-powered dyslexia testing and personalized learning paths	Good accuracy across all test types; generates personalized reading & memory exercises	English-only; does not support bilingual or non-English children
Facoetti et al. (2014)	Early identification of dyslexia through serious games	Game-based activities are effective for very young children; multiplatform support widens reach	Early detection only; no personalized remediation or bilingual support
Jency K, Lavaniya T & Swaminathan K (2025)	Two-stage parent-supervised dyslexia screening	SVM highest accuracy; parental input improves	No phonological training component; language context not specified

Author(s) & Year	Focus	Key Finding	Gap / Limitation
		assessment; stepwise approach is practical	
Thelijjagoda et al. (2019)	Mobile app to improve reading, writing, and speaking skills	Rare local Sri Lankan tool; successfully tested with children; shows local feasibility	No phonological awareness training or bilingual (Sinhala/English) sound mapping

1.3 Research Gap

When looking across the existing literature, a consistent and concerning pattern emerges — the tools that have been built to support dyslexic children simply were not built with children like those in Sri Lanka in mind.

Almost every digital phonological intervention available today, including Nexia Tutor, The Hope [6] and Lexa, was designed and tested on monolingual children, most often in Western contexts. None of them account for what it actually means to grow up learning in both Sinhala and English at the same time. When a Sri Lankan child hears a sound in one language and tries to map it to a letter in another, that is a genuinely different cognitive experience — and no existing tool acknowledges it, let alone supports it; Biize et al [2] pointed this out directly, noting that phonological tools built for one linguistic context cannot simply be transplanted into another and expected to work.

What makes this even more frustrating is that most of these tools ask children to read their instructions. For a child who already struggles with reading, being asked to read before they can even begin a reading support activity is a barrier that should never have been built in the first place. There is no tool in the literature that has been genuinely designed, from the ground up, so that a child who cannot read a single word can still participate fully and independently.

Then there is the question of what happens when a child makes a mistake. Children with dyslexia very commonly confuse letters like b and d, or p and q — not randomly, but in consistent, predictable patterns that tell us something specific about how they are processing letter orientation [6]. Despite this being well known in the field, none of the reviewed tools detect these patterns in real time, and none respond with feedback that is specific to the exact confusion the child just demonstrated. They simply mark the answer wrong and move on, which does nothing to address the underlying difficulty.

Running through all of these problems is a broader issue of tools being used in contexts they were never meant for. Biize et al [2] showed through their own data that standardized tests built for Western children give unreliable results when used elsewhere, because the words, the sounds, and the cultural references are unfamiliar.

Finally, while some studies have shown that game-based approaches can work well for young children with dyslexia, no one has yet brought together game-based phonological activities, bilingual Sinhala-English audio support and content that is actually relevant to what Sri Lankan Grade 1–3 children are learning in school — all in the same tool. These elements exist separately in different systems, but the combination that a Sri Lankan child actually needs has not been built yet.

This module is designed to address exactly that. It starts from the assumption that the child using it may not be able to read, may be hearing two languages every day, and may be consistently confusing specific pairs of letters — and it builds every design decision around those realities.

2 OBJECTIVE

2.1 Main Objective

To design and develop an audio-first, game-based Phonological Awareness and Spelling Module for Grade 1 to 3 children with dyslexia in Sri Lanka, which helps them practice phonological skills and spelling without needing to read instructions, and which detects specific letter confusion errors to provide targeted support.

2.2 Specific Objectives

1. To design a drag-and-drop sound block interface that allows children to physically pick up, move, and place letter tiles to build and manipulate words — where every tile speaks its sound aloud when touched, and every action is guided by audio rather than written text, ensuring that a child who cannot yet read a single word can still engage with the activity completely on their own.
2. To develop a set of "change one sound" puzzle activities where children listen to a word, identify one phoneme within it, and swap it out to form a new word — with audio cues playing at each step and visual feedback confirming whether the change was correct, so that the focus stays entirely on sound manipulation rather than reading ability.
3. To build a series of spelling construction activities that deliberately minimise on-screen text and instead deliver all task instructions through audio, so that children with severe reading difficulties are never blocked from participating by the very barrier the module is designed to help them overcome.
4. To implement a real-time error pattern detection system that goes beyond simply marking an answer wrong — one that recognises when a child is consistently confusing specific letter pairs such as b and d, or p and q, and responds with targeted, child-friendly feedback that addresses that specific confusion directly rather than offering a generic correction.
5. To create a Sinhala-English sound mapping feature that identifies phonemes the two languages share or closely resemble, and uses that common ground as a bridge — allowing bilingual children to approach unfamiliar English sounds through the Sinhala sounds they already know and feel comfortable with, rather than treating English phonology as something entirely foreign.

3 METHODOLOGY

This project follows a Design Science Research approach, where the goal is to design and build a useful system (module) and then evaluate how well it works. The work is split into the following stages:

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, and the reasoning behind that choice is straightforward — the goal here is not simply to study the problem but to actually build something that solves it. DSR is specifically suited to research where the intended output is a functional artefact, in this case an interactive phonological awareness and spelling module, that addresses a clearly identified real-world problem. Unlike purely observational or analytical research approaches, DSR demands that the solution be designed, built, evaluated, and refined in response to real user needs. This makes it the most honest and appropriate fit for a project where the measure of success is not just theoretical correctness but whether the tool actually works for dyslexic children in Sri Lanka.

The research will move through five structured phases drawn from the DSR framework. It begins with problem identification and motivation, where the scope of phonological awareness deficits in bilingual Sri Lankan children is established through literature and expert consultation. This feeds into the definition of objectives, which frames exactly what the module needs to achieve. The third phase is the design and development of the artefact itself, where the drag-and-drop interface, audio engine, consonant reversal detection system, and Sinhala-English phoneme mapping are iteratively built. The fourth phase is demonstration, where the module is tested with actual users in a real setting. The fifth and final phase is evaluation, where pre and post assessments measure whether the module produced genuine improvements in phonological awareness skills. Crucially, DSR does not treat these phases as a straight line — findings from evaluation feed back into design, meaning the module is continuously refined until it genuinely meets the needs it was built to address.

What makes this approach particularly well-suited to this module is that it keeps the child at the centre of every decision. Rather than designing the tool in isolation and then testing it at the end, DSR builds testing and reflection into every stage of the process. If a child cannot understand an activity, that is not a final result to be reported — it is a signal to go back and redesign. If the consonant reversal detection produces false positives, the model is retrained before the next round of testing. This cycle of designing, building, testing, and improving continues until the module is genuinely usable, accessible, and effective for the specific population it was built for. In a project where the users are young children with learning difficulties, that kind of iterative commitment to getting it right is not just methodologically sound — it is the only responsible way to work.

3.2 Technical Methods and Algorithms

- A drag-and-drop interface will be built for the sound block manipulation activities. Touch support will be included so the module works on tablets, which are commonly used in Sri Lankan schools.
- Text-to-Speech (TTS) technology will be used to read all instructions aloud in both Sinhala and English, so children do not need to read any text to use the module.
- A machine learning model (based on Support Vector Machines, as shown to work well in [1] and [5]) will be used to detect error patterns such as b/d and p/q reversals from children's responses.
- A bilingual phoneme mapping library will be created to connect similar sounds in Sinhala and English, drawing on the findings of [2] about the importance of culturally grounded tools.
- Audio feedback will be triggered in real time when a child makes an error, giving immediate and encouraging guidance.

3.3 Tools and Platforms

- React Native: will be used to build the mobile application because it produces a single codebase that runs on both Android and iOS tablets without needing to be rebuilt separately for each platform. This matters practically because Sri Lankan schools use a mix of devices, and building the module twice would significantly slow down development and make future updates harder to maintain.
- Google Text-to-Speech API: will handle the generation of all spoken instructions, word pronunciations, and phoneme audio across both Sinhala and English. Rather than relying on pre-recorded clips that are fixed and difficult to update, the API allows the module to generate natural-sounding speech dynamically, which gives far more flexibility when new words or activities are added during development and refinement.
- Python (scikit-learn): will be used to train and evaluate the error pattern detection model. Scikit-learn provides a well-tested and widely understood implementation of Support Vector Machines, which allows the model to be built, tested, and refined in a transparent and reproducible way — important for a research context where the reasoning behind the model's decisions needs to be explainable, not just accurate.
- Firebase: will be used to store each child's activity data, progress records, and session logs securely in the cloud. Its offline-first capability is particularly relevant here, as it ensures the module continues to function and record data even when internet connectivity is unreliable, and then syncs automatically once a connection is restored — a realistic requirement for many Sri Lankan school environments.
- Figma: will be used in the early stages of the project to design and prototype the user interface before a single line of code is written. Designing in Figma first allows the layouts, interactions, and visual structure of each activity to be tested and refined with real feedback from teachers and children, catching usability problems at the stage where they are cheapest and easiest to fix.

3.4 Data Collection and Ethical Considerations

Data will be collected from Grade 1 to 3 children who have been identified as having dyslexia or being at risk. All data collection will be done with the informed consent of parents and the approval of the school administration. Children's data will be stored securely and anonymized. No personally identifying information will be stored alongside learning data. Ethical clearance will be obtained from the SLIIT Ethics Committee before any data is collected.

3.5 Validation Strategy and Performance Metrics

- The accuracy of the error pattern detection model will be measured against a held-out test set of children's responses, with a target accuracy of above 90% based on the benchmark performance reported by comparable SVM-based classification models in the reviewed literature. This threshold was chosen deliberately — anything below it would mean the module is misidentifying a child's confusion too often, which could lead to incorrect feedback being delivered at a critical moment in the learning process.
- A simplified usability questionnaire will be administered to both children and their teachers after the evaluation period. For children, the questions will be visual and spoken rather than written, using smiley-face scales and audio prompts to make sure the feedback collected actually reflects their experience rather than their ability to read a form. Teacher responses will focus on how practical and reliable the module felt to use in a real classroom setting.
- Pre- and post-assessments of phonological awareness will be conducted at the start and end of a four-week evaluation period to measure whether the module produced genuine, measurable improvement in the skills it was designed to target. These assessments will cover phoneme segmentation, phoneme blending, and consonant discrimination, giving a clear picture of where each child started and how far they moved across the four weeks of use.
- Task completion rate will track the percentage of activities that children are able to finish from start to finish without needing an adult to step in and help. These metric matters because it reflects not just whether the module works technically, but whether it is genuinely accessible and understandable to a child working independently — which is one of the core design commitments of the entire module.

4 HIGH-LEVEL SYSTEM ARCHITECTURE

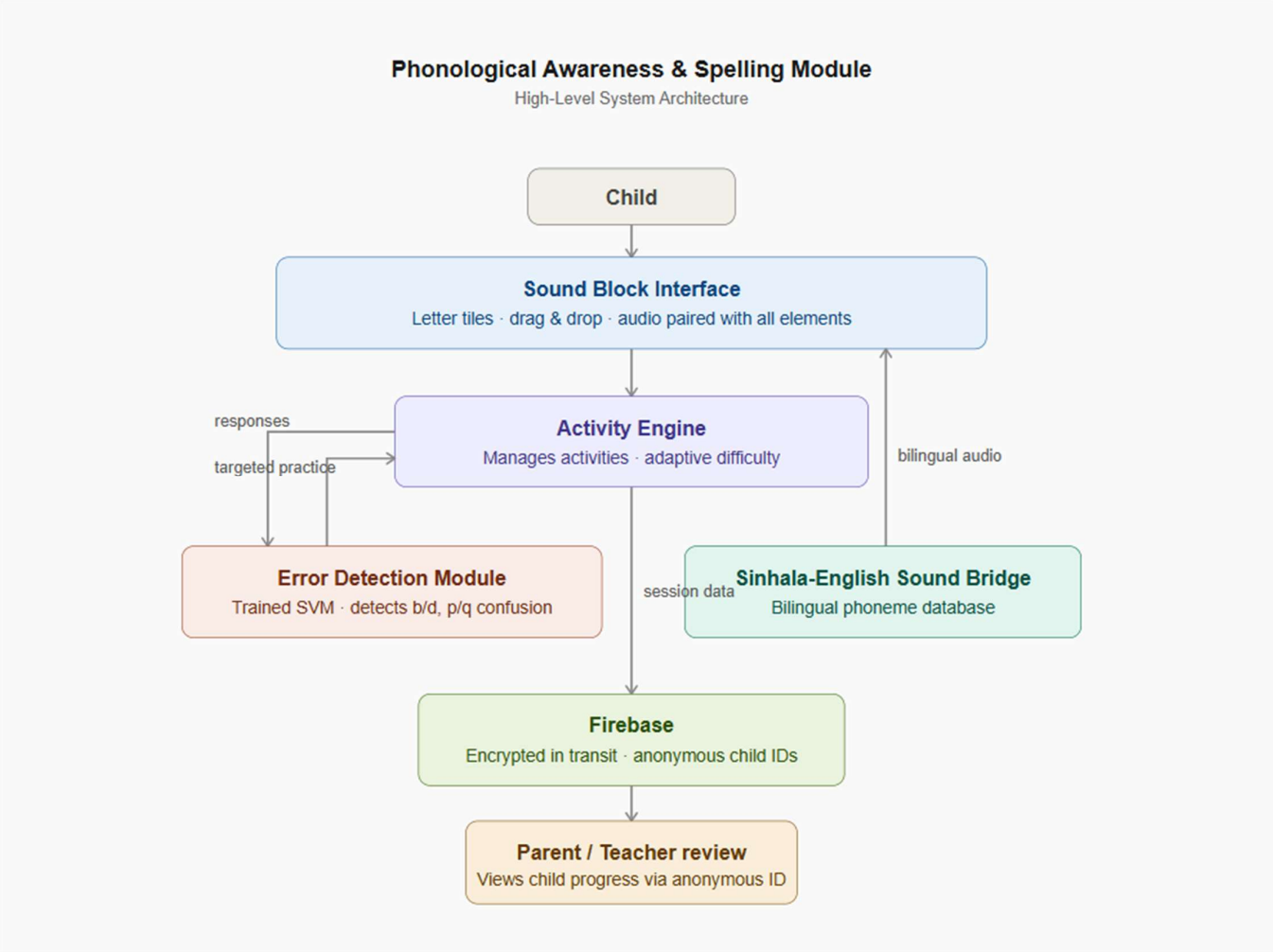
The Phonological Awareness and Spelling Module is one component of a larger, multi-module dyslexia support system. The full system also includes other modules such as Memory & Sequencing Module, Eye-Tracking Based Reading (Behaviour and Brain Processing Speed Assessment for Dyslexic Children) and Pronunciation & Speech Sound Learning. This module handles everything related to phonological awareness and spelling practice.

At a high level, the module has four main parts working together:

- Sound Block Interface: This is what the child sees and interacts with. It shows letter tiles that children can drag, drop, and rearrange. All visual elements are paired with audio so the child never has to read instructions. This part communicates with the Activity Engine.
- Activity Engine: This manages which activities are shown to the child and in what order. It knows about the three activity types: sound block manipulation, 'change one sound' puzzles, and spelling construction tasks. It adjusts the difficulty based on how the child is doing. It sends the child's responses to the Error Detection Module.
- Error Detection Module: This is the intelligent part of the system. It analyses the child's answers and looks for patterns in their mistakes, especially consonant confusion errors like b/d and p/q. It uses a trained SVM model to identify these patterns. When it finds an error pattern, it tells the Activity Engine to show targeted practice for that specific confusion.
- Sinhala-English Sound Bridge: This is a database of phonemes that are similar across both Sinhala and English. When a child is working on a sound that has a Sinhala equivalent, the system plays both the English and Sinhala version of the sound. This helps bilingual children connect what they already know in Sinhala with what they are learning in English.

Data flows from the child's touch input through the Sound Block Interface to the Activity Engine. The Activity Engine records the response, sends it to the Error Detection Module for analysis, and uses the result to decide what activity to show next. All session data is saved securely to Firebase, where a teacher or parent can later review the child's progress.

For security, all data is encrypted in transit and at rest. The system does not store the child's name alongside their learning data. Each child is identified only by a system-generated ID that is known only to the parent or teacher. The system is designed to work offline as well, so children in areas with limited internet access can still use it.


Phonological Awareness & Spelling Module
High-Level System Architecture
Child
Sound Block Interface
Letter tiles · drag & drop · audio paired with all elements
responses
targeted practice
Activity Engine
Manages activities · adaptive difficulty
bilingual audio
Error Detection Module
Trained SVM · detects b/d, p/q confusion
session data
Sinhala-English Sound Bridge
Bilingual phoneme database
Firebase
Encrypted in transit · anonymous child IDs
Parent / Teacher review
Views child progress via anonymous ID

5 USER REQUIREMENTS

5.1 Stakeholder Identification

- Primary users: Grade 1 to 3 children in Sri Lanka who have dyslexia or are at risk of dyslexia.
- Secondary users: Teachers and special education staff who monitor children's progress.
- Supporting users: Parents and guardians who help set up the app and review reports.
- System administrators: SLIIT research team who maintain and update the system.

5.2 Functional Requirements

FR	Requirement	Description
FR01	Sound Block UI	The system shall provide a drag-and-drop interface with letter tiles that children can use to build and rearrange words.
FR02	Audio Instructions	All activity instructions shall be delivered through audio in both Sinhala and English, with no reading required from the child.
FR03	Change One Sound Puzzle	The system shall provide 'change one sound' puzzles where children swap one phoneme in a word to create a new word.
FR04	Spelling Construction	The system shall offer spelling activities where children construct words using sound blocks guided by audio cues.
FR05	Error Detection	The system shall detect consonant confusion patterns (b/d, p/q reversals) from children's responses and flag them for targeted practice.
FR06	Bilingual Sound Mapping	The system shall connect similar phonemes between Sinhala and English and play both versions of a sound when introducing a new phoneme.
FR07	Progress Tracking	The system shall record each child's activity scores and error patterns and display a summary to the teacher or parent.
FR08	Adaptive Difficulty	The system shall adjust the difficulty of activities based on the child's performance over time.

5.3 Non-Functional Requirements

Category	Requirement
Usability	The interface must be usable by children aged 6 to 11 without reading any text. All interactions must be touch-based and visually clear.
Performance	Activities must load within 2 seconds. Audio must play without noticeable delay after a child's action.
Security	Children's data must be encrypted. No personally identifying information shall be stored alongside learning data.
Availability	The core activities must work offline so children can use the module without internet access.
Accessibility	Font sizes, button sizes, and color contrasts must meet child-friendly accessibility standards. Audio alternatives must exist for all visual content.
Scalability	The system must be able to support up to 500 concurrent child sessions without performance degradation.

6 COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary market is Sri Lankan schools that serve children from Grades 1 to 3. Special education units, child development centers and private therapists who work with dyslexic children are also key customers. Parents of dyslexic children who want a home-based support tool make up a secondary market.

6.2 Value Proposition

This is the first phonological awareness support tool built specifically for bilingual Sinhala-English children in Sri Lanka. It removes the barrier of reading instructions by using audio, which means even children with severe reading difficulties can use it independently. It gives teachers and parents real-time data on a child's specific error patterns, helping them target their support more effectively.

6.3 Revenue Model

- School subscription: Schools pay a yearly fee per student to access the full module.
- Freemium for parents: Parents can download a limited free version; a full version is available for a reasonable monthly fee.

6.4 Cost Estimation

- Development cost: Estimated at LKR 23,000 for the first version (developer time, tools, and testing).
- Server/cloud hosting: Approximately LKR 8,000 per year (Firebase and cloud services).
- Audio recording (Sinhala voice): One-time cost of approximately LKR 20,000.

6.5 Pricing Strategy

- School subscription: LKR 500 per student per year.
- Parent subscription (full version): LKR 200 per month.

6.6 Competitive Advantage

No other tool on the market connects Sinhala and English phonemes for dyslexic children. The audio-heavy design is unique among existing dyslexia apps. The built-in error pattern detection is more targeted than generic phonics apps.

6.7 Intellectual Property Considerations

The Sinhala-English phoneme mapping library and the error detection algorithm will be documented and may be filed as an innovation with the National Intellectual Property Office of Sri Lanka. The source code will be protected under a software copyright.

7 BUDGET AND JUSTIFICATION

Item	Estimated Cost (LKR)	Justification
Development laptop (available)	-	Required for app development and testing
Android tablet for testing (available)	-	Used to test touch interactions with children
Google Cloud TTS API	5,000	For Sinhala and English audio generation
Firebase (Blaze plan)	5,000	For secure data storage and offline sync
Sinhala voice recording (professional)	20,000 (one-time)	High-quality native Sinhala audio for children
Figma (Pro plan)	6,000	For UI/UX design and prototyping
Research and travel costs	15,000	School visits for user testing and data collection
Total (First Year)	**51,000**	

8 WORK BREAKDOWN STRUCTURE (WBS)

Task	Description	Assigned To	Timeline
T1	Literature review and gap analysis	Fernando K.H.D.M	Week 1-2
T2	UI/UX design and prototype (Figma)	Fernando K.H.D.M	Week 3-4
T3	Sound block drag-and-drop interface development	Fernando K.H.D.M	Week 5-7
T4	Change one sound puzzle development	Fernando K.H.D.M	Week 8-9
T5	Spelling construction activity development	Fernando K.H.D.M	Week 10-11
T6	Sinhala-English phoneme mapping library	Fernando K.H.D.M	Week 12-13
T7	Error pattern detection model (SVM training)	Fernando K.H.D.M	Week 14-15
T8	Integration of TTS audio and bilingual feedback	Fernando K.H.D.M	Week 16-17
T9	User testing with Grade 1-5 children	Fernando K.H.D.M	Week 18-19
T10	Evaluation, refinement, and final report	Fernando K.H.D.M	Week 20-22

9 GANTT CHART

The Gantt chart below provides a semester-wise breakdown of the project timeline. Semester 1 covers research, design, and core game development; Semester 2 covers advanced features, error detection, testing, evaluation, and documentation.

Task	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
1. Literature Review & Needs Analysis	■	■									
2. Requirements Specification		■	■								
3. Phoneme & Audio Asset Planning		■	■								
4. Game Design & Wireframing			■	■							
5. Drag-and-Drop Interface Dev.				■	■						
6. Change One Sound Puzzle					■	■					
7. Spelling Construction Activity					■	■					
8. Error Pattern Detection Engine						■	■				
9. Consonant Remediation Activity						■	■				
10. Sinhala-English Mapping Game							■	■			
11. Back-End API & DB Integration						■	■				
12. Educator Dashboard								■	■		
13. Unit & Integration Testing								■	■		
14. User Testing & Evaluation									■	■	
15. Documentation & Final Report										■	■

10 Reference

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11 APPENDICES

Appendix A: Sample Questions

The following are sample questions used in the initial phonological awareness screening that feeds into the module's adaptive engine:

1. Phoneme Identification — Voiced/Unvoiced Pair Discrimination

Listen to these two sounds: /b/ ... /d/. Are they the same sound or different sounds? (Audio plays /b/ then /d/ with a pause between) I will say a word. You drag the matching letter tile. Ready?

2. Word-Initial Sound Identification

Listen to the word: "bed". What is the first sound you hear? (Audio plays: "bed" — child selects from tiles: b / d / p / q) Which two words sound the same at the end? 'cat', 'dog', 'hat'.

Appendix B: Preliminary UI Mockup Description

The main screen shows a row of colorful letter tiles at the bottom. A large picture of an object (for example, a fish) is shown in the center of the screen. A friendly audio prompt says: 'Drag the letters to spell the word.' The child drags and drops the tiles into a target area. Correct placements are highlighted in green with a happy sound effect. Incorrect placements gently animate back to their original position with an audio cue saying 'Try again!'

The Sinhala-English bridge mode shows two flags (Sri Lanka and UK) and plays both the Sinhala and English version of the target phoneme when the child taps a special audio button.

Appendix C: Risk Analysis

Risk	Level	Mitigation
Difficulty getting access to dyslexic children for testing	Medium	Work with SLIIT contacts and local schools early to arrange testing sessions well in advance.

Sinhala TTS quality being too robotic for young children	Medium	Use a professional Sinhala voice actor to record key phoneme sounds instead of relying purely on TTS.
Children finding the activities too hard or not engaging	High	Conduct usability testing with 5 children in an early prototype stage and adjust before full development.
SVM error detection model having low accuracy on local data	Medium	Collect enough local training data (at least 200 sessions) before training the final model.
Internet unavailability in some schools	High	Design the core module to work fully offline, with data syncing only when a connection is available.

Appendix D: Ethical Clearance

This project will be submitted for ethical review to the SLIIT Research Ethics Committee before any data is collected from children. All participating schools will provide written consent. Parents will be given a plain-language information sheet explaining the study and asked to sign a consent form. Children will be told in simple terms what the activities involve and will be free to stop at any time. No images or recordings of children will be made. Only activity score data will be stored.